

☒ Easy Level (10 MCQs)

1.

What is the purpose of a patch in Linux?

- A) Upgrade hardware
- B) Repair filesystem
- C) Fix bugs or add features
- D) Reboot system

Ans: C

2.

Which tool applies patch files to source code?

- A) apply
- B) git
- C) patch
- D) rpm

Ans: C

3.

What is the default graphical system in Linux?

- A) bash
- B) Wayland
- C) X server
- D) gdm

Ans: C

4.

Which command starts a graphical X session?

- A) gnome-start

- B) xinit
- C) startx
- D) startxserver

Ans: C

5.

Which command lists running processes?

- A) listproc
- B) jobs
- C) top
- D) procview

Ans: C

6.

Where are most system logs stored in Linux?

- A) /boot/
- B) /home/logs
- C) /etc/log
- D) /var/log

Ans: D

7.

Which tool applies live kernel patches without reboot?

- A) rpm
- B) apt
- C) kpatch
- D) patch

Ans: C

8.

Which file contains the system-wide X server config?

- A) /etc/xorg
- B) /etc/X11/xorg.conf
- C) /usr/bin/Xorg
- D) /etc/X11/Xdefault

Ans: B

9.

Which utility can update system packages in Debian?

- A) yum
- B) zypper
- C) apt
- D) rpm

Ans: C

10.

Which display manager is used by GNOME by default?

- A) GDM
- B) XDM
- C) SDDM
- D) LightDM

Ans: A

☒ **Medium Level (20 MCQs)**

11.

Which command creates a patch file from code differences?

- A) mkpatch
- B) diff -u
- C) patchgen
- D) createpatch

Ans: B

12.

Where are scheduled jobs defined in Linux?

- A) /etc/cron.conf
- B) /etc/crontab
- C) /etc/schedule
- D) /var/cron

Ans: B

13.

Which command checks memory usage?

- A) memcheck
- B) meminfo
- C) free -h
- D) htop

Ans: C

14.

What is the role of the X server?

- A) Run background services
- B) Provide terminal access
- C) Handle GUI input/output

D) Install updates

Ans: C

15.

What does the xinput command do?

A) Display screen resolution

B) Configure input devices

C) Show memory info

D) Manage USBs

Ans: B

16.

Which package provides the base X server in most distros?

A) xinit

B) xserver-xorg

C) x11-common

D) xorg-dev

Ans: B

17.

What does patch -p1 < file.patch do?

A) Compress the file

B) Revert patch

C) Apply patch with one level path strip

D) Skip file

Ans: C

18.

Where can you persist kernel parameters?

- A) /etc/sysctl.conf
- B) /etc/modprobe.d/
- C) /boot/grub.cfg
- D) /etc/default/kernel

Ans: A

19.

Which utility is used for package updates in RHEL/CentOS?

- A) apt
- B) yum/dnf
- C) pacman
- D) zypper

Ans: B

20.

How to make a patch from two directories?

- A) patch dir1 dir2
- B) diff -urN dir1 dir2 > update.patch
- C) diff -f dir1 dir2
- D) cmp dir1 dir2

Ans: B

21.

Which X utility shows OpenGL support?

- A) glxinfo
- B) xrandr
- C) lshw
- D) xdpinfo

Ans: A

22.

Which file contains per-user X session configs?

- A) ~/.Xresources
- B) ~/.bashrc
- C) ~/.xinitrc
- D) Both A and C

Ans: D

23.

Which tool can reboot the system safely?

- A) reboot
- B) shutdown -r
- C) systemctl reboot
- D) All of the above

Ans: D

24.

Which package manager uses .rpm format?

- A) apt
- B) pacman
- C) rpm/yum
- D) dpkg

Ans: C

25.

Which scheduler runs once at a future time?

- A) cron
- B) systemd
- C) at

D) timer

Ans: C

26.

Where is the initrd image typically stored?

A) /boot/initrd.img-*

B) /etc/init.d/

C) /usr/share/init/

D) /dev/initramfs

Ans: A

27.

Which X command helps change screen resolution?

A) xrandr

B) xvid

C) displayctl

D) xinput

Ans: A

28.

Which command shows currently loaded kernel modules?

A) lsmod

B) modlist

C) dmesg

D) insmod

Ans: A

29.

Which command installs packages on Fedora?

A) pacman

B) zypper

C) dnf

D) dpkg

Ans: C

30.

Which file lists automatic updates in Debian?

A) /etc/apt/update.conf

B) /etc/apt/apt.conf.d/20auto-upgrades

C) /etc/system/update.conf

D) /etc/updates.conf

Ans: B

☒ Hard Level (10 MCQs)

31.

Which of the following is used for live patching without reboot?

A) apt live

B) kpatch

C) patch -L

D) yum --live

Ans: B

32.

To patch kernel source code manually, which directory is used?

- A) /lib/modules/
- B) /usr/src/linux
- C) /etc/kernel
- D) /boot/src/

Ans: B

33.

Which systemctl command lists failed units?

- A) systemctl --failed
- B) systemctl list-failed
- C) systemctl list-units --failed
- D) systemctl failed-units

Ans: A

34.

What does diff -u produce?

- A) Unified diff for patching
- B) Context diff
- C) Line-by-line diff
- D) JSON diff

Ans: A

35.

Which tool automatically applies security patches on Ubuntu?

- A) auto-apt
- B) unattended-upgrades
- C) apt-updater
- D) upgrade-guard

Ans: B

36.

Which tool loads kernel modules dynamically during boot?

- A) modprobe
- B) insmod
- C) kernel-loader
- D) depmod

Ans: A

37.

How does Wayland differ from X11?

- A) Uses more RAM
- B) Better performance & security
- C) Supports only GNOME
- D) Only works with proprietary drivers

Ans: B

38.

Which command shows CPU usage in real time?

- A) vmstat
- B) free
- C) top
- D) uptime

Ans: C

39.

Where are unit files stored in systemd?

- A) /etc/systemd/system/
- B) /usr/lib/systemd/system/
- C) /lib/systemd/system/

D) All of the above

Ans: D

40.

Which command verifies an RPM package file?

A) rpm -v

B) rpm -qf

C) rpm -V

D) rpm -i

Ans: C